Arnav Harshit

Fourth Year Undergraduate
Department Chemical Engineering
Indian Institute of Technology Kanpur

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Examination	Degree/Certificate	Institute	Year	CPI/%
Graduation	B.Tech	Indian Institute of Technology Kanpur	2023–Present	**7.94**/10
Intermediate/+2	CBSE(XII)	BSN Academy, Sr. Sec School, Kota	2023	90.60%
Matriculation	CBSE(X)	BSN Academy, Sr. Sec School, Kota	2021	94.80%

Work Experience

UnifyApps *May'26 – Jul'26*
Product Engineer Intern **Pre-Placement Offer (PPO) Received**

- Developed the **Agent Optimizer**, a conversational AI agent that diagnoses **25+ production AI agents** via **trace analysis** across cost, latency, and instruction quality
- Orchestrated **6 composable automations** via a **parallel fan-out wrapper**, cutting batch analysis from linear to **near-constant**
- Isolated **high-token outliers** varying **30x per task** through **SQL-driven case selection** over the analytics warehouse
- Condensed **45 anti-pattern rules** into a diagnostic taxonomy, evaluating system prompts to trace root causes of **LLM errors**
- Built an automated **prompt optimization framework** emitting **evidence-backed rewrites**, raising **token efficiency** by removing **redundant file-access loops**

OpusLeaf Technologies *Jun'25 – Jul'25*
Software Engineering Intern

- Designed and implemented a **backend** service aggregating and standardizing **vulnerability** data from security tools
- Developed **Python** scripts for **API integration** (e.g., **SonarQube**) with authentication and pagination handling
- Delivered groundwork for **API/UI integration**, improving **backend reliability, scalability, and extensibility**

Invictron Systems Pvt. Ltd *May'25 – Jul'25*
Software Development Engineer

- Designed a peer-to-peer messaging app enabling **offline chat** and **location sharing** without any server
- Built a WhatsApp-style **native Android** app with Bluetooth/Wi-Fi chat, device discovery, and **offline maps**
- Achieved stable messaging across **10+** devices with offline location tracking and direct relay messaging

Programming Achievements

- Achieved a highest rating of **1718 (Expert)** on the **Codeforces** competitive programming platform DangerWizz
- Secured global ranks **188** and **115** in **Codeforces Round #1115** and **Educational #193**, among **28,000+** participants

Scholastic Achievements

- **Finalist (Top 75 teams)** among thousands at **Code with Cisco 2026**, earning a **sponsored** trip to the finale
- Qualified **Amazon ML Summer School 2026**, among the **top 3,000** selected from **1.34 lakh+ applicants** across India
- Qualified **Stage II** of the **National Talent Search Examination (NTSE)**, among **5,000 candidates** nationwide in 2021
- Secured **International Rank 36** and **Zonal Rank 5** in the **SOF International Mathematics Olympiad**, 2020

Technical Skills

- **Languages & Core:** C | C++ | Python | Java | JavaScript | SQL | MATLAB | Linux | Git | DSA | OOP | DBMS | CN | OS
- **Web, Backend & Systems:** React.js | Node.js | Express.js | Spring Boot | REST APIs | PostgreSQL/MongoDB | Docker | AWS | Microservices | Multithreading | Parallel Computing
- **ML & AI:** NumPy | Pandas | Scikit-learn | PyTorch/TensorFlow/Keras | OpenCV | Graph Algorithms

Key Projects

Raft-Based Distributed Key-Value Store
Self Project Jul'26 – Oct'26

- Built a fault-tolerant distributed key-value store on **Raft**: leader election, log replication, heartbeats, and quorum commits
- Implemented asynchronous networking with **Boost.Asio** and a **Redis-compatible** client interface
- Validated consistency across node failures and restarts via persistent **log/WAL** recovery and multi-node testing
- Benchmarked concurrent Redis workloads, achieving **73K+ SET** and **105K+ GET requests/sec** in the provided configuration

MiniDB: SQL Database Engine
Self Project Dec'25 – Feb'26

- Built a relational SQL engine with **query parsing**, **AST generation**, schema management, and execution
- Implemented disk storage with **slotted pages**, buffer management, serialization, and **B+ tree indexing**
- Developed a **Volcano-style** execution pipeline for scans, filtering, and projection, with modular **CMake** builds and **unit testing**

FEDAR+: Federated Appliance Recognition
Self Project Sep'25 – Nov'25

- Built a **PyTorch** federated learning pipeline recognizing appliances from **1D power-usage footprints** on **UK-DALE** and **Tracebase**, reaching **95%** accuracy
- Corrected **model weights and label distributions** via **KL-divergence**, holding >**88%** accuracy at **30%** label noise
- Implemented **Dirichlet** partitioning across **20 non-IID** clients and **label-flip** evaluation, gaining **15–27%** over an **LSTM** baseline

ShiftGuard-10: Robust Image Classification
EE708 Course Project | Prof: Rajesh M. Hegde Jan'26 – Apr'26

- Built a **WideResNet-28-6** classifier in **PyTorch** for **10-class image classification** under distribution shift
- Applied **RandAugment, Cutout, Mixup/CutMix** and severe-class-imbalance handling to improve generalization
- Ensembled two diverse models with **15-pass TTA**, achieving **88.3% Macro F1** vs. a **67.3% baseline**

Research Experience

Small-World Properties for Enhanced GCN Performance
Undergraduate Project (UGP) | Prof: Rajesh M. Hegde Dec'24 – Apr'25

- Integrated **small-world links** into graphs via **top-k feature similarity** and **entropy-centrality filters** to strengthen topology
- Optimized **hyperparameters** (α, **k, threshold**) through **Small-World Optimization** for robust, data-driven rewiring
- Achieved up to **10% higher accuracy** across datasets without altering the **GCN architecture**, demonstrating model robustness

Positions Of Responsibility

GDG Trainee, Google Developer Group *Oct'24 – Apr'25*

- Initiated 6+ **tech projects** and curated content to enhance learning in **ML, Web, and Cloud** through peer-led efforts

Raktarpan Secretary, IIT Kanpur *Aug'24 – Apr'25*

- Enabled collection of over **720+ blood units** through **8 camps**, positively impacting an estimated **2400+ lives**

Relevant Courses

(† A* Grade | * Online)

Data Structures and Algorithms
Machine Learning (Amazon ML)*
Generative AI and ChatGPT*
Fund. of Data Science & MI†
Machine Learning (Coursera)*
Fundamentals of Computing
Linear Algebra and ODE
AWS Cloud Practitioner*
Introduction to Electronics